

Accurate Age, Uncertain Utility

A Critical Analysis of Chronological-Age Prediction, Biological Interpretation, and Clinical Evidence in a Proteomic Aging Clock

A critical analysis of “Proteomic aging clock predicts mortality and risk of common age-related diseases in diverse populations”

Argentieri et al. (2024), Nature Medicine, 30, 2450–2460

Sheryl S.

Research Portfolio Series · Critical Analysis #03

September 2026

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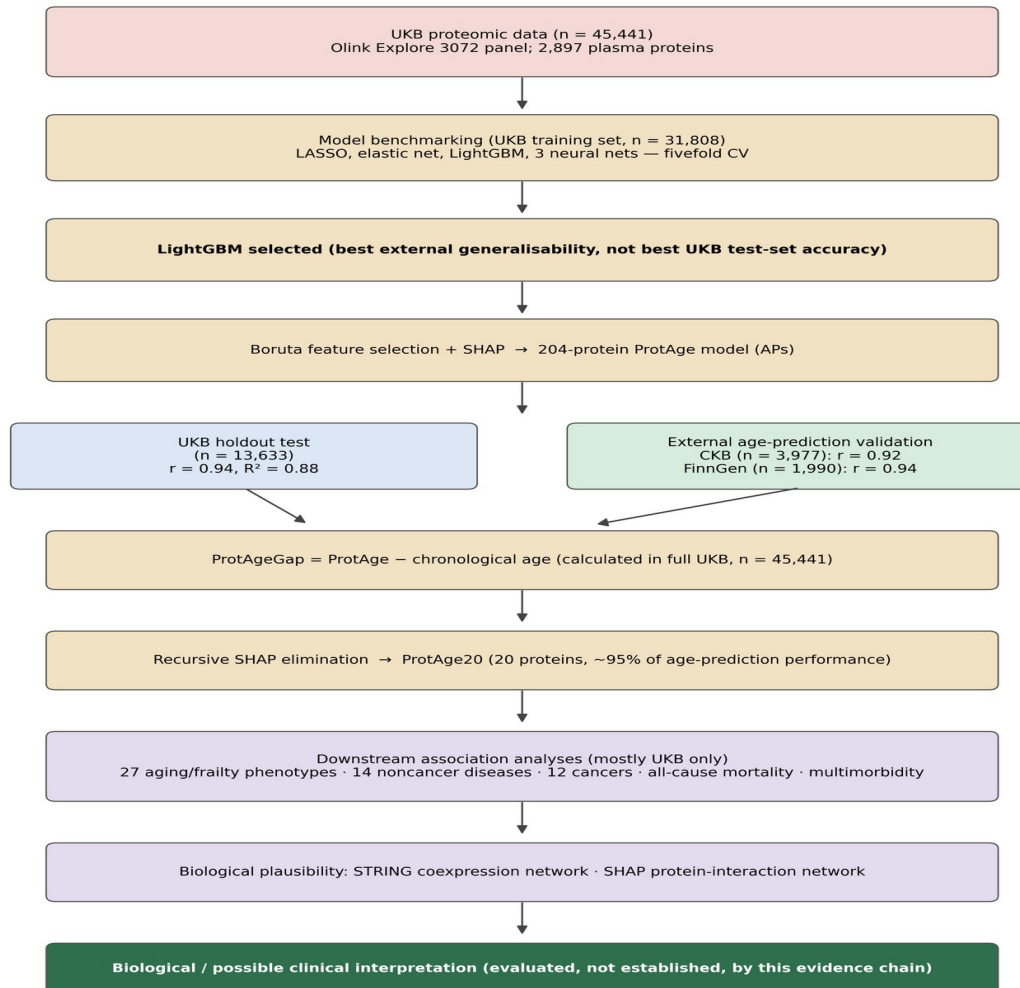
1. Central Claim and Evidentiary Overview

Argentieri et al. (2024) develop a proteomic aging clock in the UK Biobank (UKB) using plasma expression of 2,897 proteins measured on the Olink Explore 3072 platform, and then ask how far that clock can be pushed: from predicting chronological age, to describing biological aging, to flagging disease and mortality risk. The central claim, as I read it, is that a protein-based age gap — ProtAgeGap, the difference between protein-predicted age (ProtAge) and chronological age — captures information about aging that is reproducible across genetically and geographically distinct populations and that associates with a very wide range of downstream health outcomes.

The evidence chain runs: UKB proteomic data → chronological-age model training (six machine-learning methods benchmarked) → Boruta/SHAP feature selection to 204 proteins → UKB holdout testing → external age-prediction validation in the China Kadoorie Biobank (CKB) and FinnGen → calculation of ProtAgeGap in the full UKB → associations with 27 aging-related phenotypes, 14 noncancer diseases, 12 cancers, multimorbidity, and all-cause mortality → biological interpretation via SHAP importance and a high-confidence STRING coexpression network. Each link in that chain licenses a different strength of claim, and the paper does not provide equally strong evidence for all of them.

Model-development evidence is strong: the gradient-boosted model (LightGBM) explained most of the variance in chronological age in the UKB holdout test set. External transportability evidence is also strong for the age-prediction task specifically, since the same trained model reproduced similarly high correlations in CKB and FinnGen. Epidemiological association evidence is extensive — ProtAgeGap was tested against dozens of outcomes with consistent, FDR-corrected, dose-graded results. Biological plausibility evidence is present but indirect, resting on feature-importance measures and database-level protein-interaction enrichment rather than on functional experiments. Evidence relevant to clinical prediction — whether ProtAgeGap actually improves risk stratification for an individual patient beyond what age and standard risk factors already provide — is the thinnest link in the chain, because the paper does not report a head-to-head comparison of a model with and without ProtAgeGap.

These are related questions, but they are not interchangeable, and treating them as though they were would overstate what the study shows. The rest of this analysis works through the chain roughly in that order: what the clock was built to do, how convincingly it does it, and where the evidence runs out before reaching claims about biological mechanism or bedside utility.



Conceptual diagram constructed by the author; not reproduced from Argentieri et al. (2024).

Figure 1. From UKB proteomic data to biological and clinical interpretation. Conceptual diagram constructed by the author; not reproduced from Argentieri et al. (2024).

2. Theoretical Foundation

This section covers only what I need to follow the critique in Section 4 — it is not a general primer on aging biomarkers or gradient boosting.

2.1 Chronological Age, Biological Age, and an Aging Clock

Chronological age is simply time since birth. Biological aging is a broader, less precisely defined construct: the idea that people of the same chronological age can differ in how much physiological decline they have accumulated, and that this decline is what actually drives disease risk and mortality. An ‘aging clock’ is any model that uses biological data (DNA methylation, proteins, metabolites) to produce a predicted age, on the logic that deviations between predicted and chronological age might track this hidden variation in biological wear.

Protein-predicted age (ProtAge) in this paper is the output of a model trained to predict chronological age from plasma protein levels. ProtAgeGap is ProtAge minus chronological age. The important conceptual point, which I did not fully appreciate before working through the Methods, is that accurate chronological-age prediction and validated measurement of biological aging are not the same achievement. A model can recover chronological age very well simply because many proteins change with age in a fairly regular way, without that residual necessarily corresponding to one coherent, causally meaningful biological-aging process. Biological age also does not have a single agreed ground truth against which any clock can be definitively validated — different tissues, systems, and omics layers can each suggest somewhat different ‘biological ages’ for the same person, so no clock can be checked against a gold standard in the way a diagnostic test can be checked against confirmed disease status.

2.2 LightGBM and Nonlinear Prediction

LightGBM is a gradient-boosted decision-tree algorithm: it builds an ensemble of shallow trees sequentially, with each new tree fitting the residual error of the ensemble so far. This lets it capture nonlinear relationships between individual proteins and age, as well as interactions between proteins, without the analyst having to specify those interactions in advance the way a linear or LASSO model would require. That flexibility is likely part of why it generalized better across cohorts than the simpler linear methods in this paper's benchmarking (Section 5).

The trade-off is interpretability. A tree-ensemble model does not hand you a simple equation the way a linear regression does; its predictions have to be explained after the fact using tools like SHAP. Strong predictive performance from a model like this tells you the model found a useful mapping from proteins to age — it does not by itself tell you that mapping corresponds to biologically interpretable structure, which is why the authors needed a separate interpretation layer (Boruta and SHAP) on top of the prediction task.

2.3 Boruta Feature Selection and SHAP

Boruta is a wrapper feature-selection method. It creates ‘shadow features’ — randomly permuted copies of each real feature, which by construction carry no true signal — and then keeps only the real features that outperform the best of these shadow features across repeated model runs. In this paper, Boruta was run with 200 trials and a 100% threshold, meaning a real protein had to beat all of its shadow-feature comparisons to be retained, which is how the authors narrowed 2,897 proteins down to 204.

SHAP (SHapley Additive exPlanations) values quantify, for a trained model, how much each feature's value pushed an individual prediction away from the model's average output, based on Shapley values from cooperative game theory. A protein with a high mean absolute SHAP value contributes substantially to the model's age predictions across participants. That is a statement about the fitted model — it describes which inputs the model leans on to produce its output. It is not, on its own, a statement about which proteins cause aging or age-related disease in the body. Two proteins can have similar SHAP importance for very different biological reasons, including reasons unrelated to aging mechanism (for example, a protein that is simply a stable, easily measured correlate of age). Similarly, SHAP interaction values describe how pairs of features jointly shape the model's predictions; they are a property of the fitted function, not a direct readout of physical protein-protein interaction, even when the resulting network is compared against a resource like STRING, which itself provides coexpression-based context rather than confirmed physical interaction for most entries.

2.4 Performance, Association, and Clinical Utility

A cluster of related but distinct statistical concepts recurs throughout this paper, and keeping them separate is central to the critique in Section 4:

- **Pearson correlation (r)** and **R^2** describe how closely predicted age tracks chronological age across a sample — they are age-prediction accuracy metrics, not measures of clinical usefulness.
- **RMSE and MAE** describe the typical size of the prediction error in years, giving a more interpretable sense of scale than r or R^2 alone.
- **Hazard ratios (HRs)** from Cox models describe the average relative increase in event rate per unit increase in a predictor, adjusted for the covariates in that model — they describe association in a defined sample, not prediction for an individual.
- **Calibration** asks whether predicted risks match observed event rates, which is a separate question from whether a score is associated with an outcome at all.
- **Discrimination** (for example, a C-index or time-dependent AUC) asks how well a score ranks people by risk.
- **Incremental prediction** asks whether adding a new marker to an existing model improves discrimination, calibration, or reclassification — a comparison, not a single model's fit statistic.
- **Clinical usefulness** is a further question again: whether using the marker would change a decision or outcome in a way that is worth the cost and effort of measuring it.

A statistically significant, FDR-corrected hazard ratio for ProtAgeGap and a disease outcome is genuine evidence that the two are associated after adjustment. It is not, by itself, evidence that adding ProtAgeGap to a clinical risk model would measurably improve that model's ability to predict who develops the disease. That distinction is the hinge on which most of Section 4 turns.

3. Data and Study Design

The development cohort is a randomly selected proteomics subset of the UKB cohort with Olink Explore data at baseline ($n = 45,441$; 54.1% female; age range 39–71 years; mean age 57, SD 8.2), after excluding two specific proteomics batches (batches 0 and 7) as described in the Methods. The authors cite prior work indicating this subset is broadly representative of the wider UKB population, but this should not be read as a nationally representative sample of the general UK population — the UKB itself is a volunteer cohort, a point I return to in Section 7. This UKB sample was split 70:30 into a training set ($n = 31,808$) and a holdout test set ($n = 13,633$).

Two independent cohorts were used for validation. CKB contributed 3,977 participants drawn from a nested case-cohort study of ischemic heart disease (54% female, age range 30–78, mean age 57, SD 12); this design matters because CKB was a nested ischemic-heart-disease case-cohort sample rather than a general cohort sample. Unlike the randomly selected proteomics subset within UK Biobank, CKB was enriched for ischemic-heart-disease cases and controls selected for that purpose. FinnGen contributed 1,990 participants (52% female, age range 19–78, mean age 56, SD 15) described as mostly healthy blood donors, with only 22 deaths (1.1%) recorded during follow-up. The Results summarise approximately 11–16 years of follow-up overall in the UKB and 11–14 years in CKB, whereas the Methods report approximately 12–16 years specifically for UKB mortality linkage; these may reflect related but not identical follow-up definitions rather than a clear error, so I note both figures here rather than picking one silently.

Proteomic profiling used the Olink Explore 3072 platform across four panels (Cardiometabolic, Inflammation, Neurology, Oncology). After restricting to proteins measured in all three cohorts and excluding three further proteins missing in over 10% of the UKB sample, 2,897 proteins remained for analysis. Missing protein values were imputed in the UKB and FinnGen using the miceforest package; CKB required no protein imputation. Six machine-learning approaches were benchmarked on the UKB training data with fivefold cross-validation (LASSO, elastic net, LightGBM, and three neural-network architectures), and LightGBM was selected based on

its external-cohort performance rather than its UKB test-set accuracy — a detail I come back to in Section 5. Boruta feature selection (using SHAP-based importance and shadow features) reduced the protein set to 204 age-predictive proteins (APs); a separate SHAP-based recursive feature elimination further reduced this to a 20-protein model, ProtAge20.

Downstream, the paper reports Cox proportional-hazards models (for mortality, 14 noncancer diseases, and 12 cancers), which used three successive levels of covariate adjustment, and linear/logistic regression models (for 27 aging-related phenotypes), which used a single specified set of demographic, lifestyle, and technical covariates (age, sex, Townsend deprivation index, assessment center, ethnicity, IPAQ activity group, and smoking status); Benjamini-Hochberg FDR correction was applied within each outcome family. Sensitivity analyses are reported separately for these two analysis families: the phenotype/biomarker models were additionally restricted to participants with no lifetime diagnosis of any of the 26 diseases studied (to reduce reverse-causation concerns), while the Cox mortality and incident-disease models were additionally restricted to never-smokers and to a normal BMI range.

It is important not to flatten this into ‘the model was validated across three diverse cohorts and all analyses generalized.’ The Figure 1 legend states explicitly that most association analyses were carried out only in the UKB, because of CKB's smaller sample size and the near-absence of incident disease in FinnGen. CKB contributes descriptive cumulative-incidence comparisons across ProtAgeGap deciles for diseases with more than ten incident cases, not the full battery of adjusted Cox models run in the UKB. FinnGen contributes only to the age-prediction validation step. Keeping development, UKB holdout testing, external age-prediction validation, and downstream UKB-centred association analyses as four analytically distinct stages is, I think, the single most important bookkeeping task for reading this paper accurately.

4. From Accurate Age Prediction to “Biological Age” — The Central Critique

This is the section where I want to be most careful, because it is easy to either overclaim (treat ProtAgeGap as a validated biological-age measure because it predicts outcomes) or underclaim (dismiss it because it was trained on chronological age). Neither move is well supported by what the paper actually reports.

4.1 What the Clock Was Trained to Predict

The supervised target for every model in the benchmarking stage was chronological age at recruitment, not any independent biological-aging measure. Recovering chronological age this accurately from 204 (or even 2,897) proteins — $r = 0.94$ in the UKB test set, and comparably high in two cohorts the model had never seen — is a genuinely strong technical result. It tells us that plasma protein levels carry a robust, cross-population signal that tracks age closely. What it does not automatically tell us is that the residual left over after this prediction (ProtAgeGap) represents a single coherent biological-aging process, as opposed to simply capturing measurement noise, cohort-specific protein-assay characteristics, or a mixture of several unrelated processes that happen to correlate with age.

I don't think this is a reason to dismiss the chronological-age training as a flaw — predicting age accurately from a fixed, well-characterized target is what makes the model reproducible and comparable across cohorts in the first place, and the paper's subsequent step of testing ProtAgeGap against dozens of independent outcomes is exactly the kind of follow-up that can strengthen a chronological-age-trained clock's claim to biological relevance. My point is narrower: the age-prediction accuracy numbers and the ProtAgeGap outcome associations are two different pieces of evidence, and only the second speaks directly to whether ProtAgeGap means anything beyond ‘how old this person's proteome looks.’

4.2 Interpreting ProtAgeGap

ProtAgeGap is calculated as ProtAge minus chronological age, so it functions as an age-prediction residual by construction. Because chronological age is used both to train the model and to define the residual, adjusting all downstream association models for chronological age is a necessary step, and the paper does this consistently. What I looked for next was evidence about the residual's own behavior: does ProtAgeGap sit at a similar scale across the age range, or does it drift for younger versus older participants (a pattern sometimes called regression toward the mean, where predictions for extreme ages get pulled toward the sample average)?

The paper reports that model error was similar across different age groups (referenced as a supplementary analysis) and that ProtAgeGap showed similar distributions by sex, by self-reported ethnicity in the UKB, and by geographic region in CKB (Fig. 2g-i). That is reassuring at the level of these specific subgroups. What I could not find in the material reviewed was a residual-versus-age scatterplot, a formal calibration slope, or an explicit statement addressing regression-to-the-mean directly. Given the reported similar error across age groups, I don't think there is positive evidence of age-dependent bias here — but I also don't think the absence of an explicit calibration-slope analysis should be silently treated as ruling the concern out. This is the kind of diagnostic that Extension 2 in Section 12 is built around.

4.3 Association Is Not Incremental Prediction

This is where I think the paper's evidence is strongest on one specific claim and least direct on an adjacent, easily conflated claim. The Cox models show that ProtAgeGap is associated with mortality and 13 of 14 noncancer diseases after adjustment for chronological age, sex, and (in the fully adjusted model) sociodemographic, lifestyle, BMI, and hypertension covariates, with hazard ratios as high as 1.16 per year of ProtAgeGap for Alzheimer's disease. That is a well-powered, FDR-corrected, dose-graded adjusted association — genuinely strong evidence that ProtAgeGap is not simply redundant with chronological age and the other covariates included.

What this does not establish is a quantified improvement in individual-level risk prediction. Demonstrating incremental value would require comparing a baseline model containing chronological age and established risk factors against the same model plus ProtAgeGap, and reporting the change in a discrimination metric (C-index or time-dependent AUC), a calibration assessment, or a reclassification statistic. From the material reviewed, I did not find this specific comparison reported for any outcome. The paper instead reports the fold-increase in cumulative incidence between the top and bottom of the ProtAgeGap distribution (for example, roughly 2.6-fold higher AD risk in the top 5% versus those with a ProtAgeGap of zero, and 5.8-fold versus the bottom 5%), which is a vivid and useful way to communicate the size of the association, but it is still a statement about group-level risk gradients, not about how much a discrimination metric moves when ProtAgeGap is added to an existing clinical model. I want to be precise about what I am and am not saying: the paper does not claim ProtAgeGap has no predictive value, and I am not saying that either. What I can say is that the paper demonstrates adjusted association convincingly and does not, in the sections reviewed, directly quantify incremental predictive performance over existing risk factors.

4.4 Model Importance Is Not Biological Mechanism

The paper identifies Boruta-selected proteins, ranks them by SHAP importance, builds a SHAP-based protein-interaction network from the trained model, and separately checks the 204 APs against the STRING database using high-confidence coexpression data, finding one significant GO biological process (anatomical structure development) and a 66-protein interconnected STRING coexpression subnetwork centered on proteins including EGFR, CXCL12, ITGAV, CXCL9, and CD8A. GDF15 stood out functionally, associating with 16 of 18 ProtAgeGap20-linked diseases.

Each of these pieces of evidence does a different job, and it is worth being explicit about which. Boruta and SHAP importance describe which proteins the fitted LightGBM model relies on to predict age — a statement about the model. SHAP interaction values describe how pairs of proteins jointly shape the model's predictions — also a statement about the model, not a direct measurement of physical protein–protein interaction. The STRING analysis provided external, database-derived coexpression context for the selected proteins, using high-confidence relationships (confidence >0.7); it did not demonstrate direct physical protein interactions or experimentally validate the proposed network within this study, and STRING membership does not, on its own, establish that a given protein causally drives any of the diseases ProtAgeGap is associated with. None of these three lines of evidence, individually or combined, constitutes a perturbation experiment, a Mendelian randomization analysis, or a longitudinal protein-change study that would more directly support a causal claim.



Figure 2. An evidence ladder for the paper's central claim. Levels 1 and 2 are strongly supported by the reported analyses; level 3 (incremental prediction) is not directly tested; level 4 (mechanism) has plausibility support only; level 5 (clinical usefulness) is untested. Conceptual diagram constructed by the author; not reproduced from Argentieri et al. (2024).

5. Model Development and Feature Selection

Six machine-learning approaches were compared for predicting chronological age from the full 2,897-protein panel: LASSO, elastic net, LightGBM, and three neural-network architectures (a multilayer perceptron, ResNet,

and TabR), each tuned via fivefold cross-validation with Optuna hyperparameter search. LightGBM showed the second-best accuracy in the UKB holdout test set but the best accuracy in the independent CKB and FinnGen samples, and the authors selected it on this basis. I think this is a defensible and, honestly, a fairly disciplined choice: optimizing model selection for external generalisability rather than in-sample UKB accuracy is exactly the kind of decision that reduces the risk of a model that looks impressive in its home cohort but degrades elsewhere. It does mean that whichever model had the single best UKB test-set number was not used, which is worth stating plainly rather than letting the headline $r = 0.94$ obscure the fact that another model scored marginally higher on that one metric.

Feature selection proceeded in two stages. First, Boruta (200 trials, 100% threshold) selected 204 proteins from the full panel using SHAP-based importance against permuted shadow features, and the LightGBM model was re-tuned on this reduced set. Second, a SHAP-based recursive feature elimination procedure, starting from the 204 Boruta-selected proteins and iteratively removing the least important one (by cross-validated mean absolute SHAP value), identified 20 proteins retaining about 95% of the 204-protein model's age-prediction performance ($r = 0.89$, $R^2 = 0.78$), with performance dropping sharply below 20 proteins.

On separation of stages: the Methods describe hyperparameter tuning via fivefold cross-validation on the training data, followed by Boruta selection, followed by re-tuning of a new model on the selected features, with both the before- and after-selection models checked for overfitting and evaluated against the same held-out UKB test set. This is a defensible pipeline structure, but I want to flag precisely what I can and cannot verify about it from the material reviewed. I do not have a per-fold breakdown showing that Boruta's shadow-feature comparisons were generated fresh within each cross-validation fold used for final evaluation, as opposed to being run once on the full training set before the final holdout evaluation; the Methods describe the sequence of steps but I could not confirm the exact nesting relationship between the Boruta step and the fivefold cross-validation loop used for the final reported test-set metrics. I am not alleging data leakage — the authors explicitly checked for overfitting and the model's performance held up in two completely external cohorts, which is a strong practical argument against a leakage explanation for the headline result. Put more simply, the exact nesting of feature selection within cross-validation is a detail I could not fully reconstruct, and I would want to see it made explicit rather than assume it either way.

A separate question is feature-selection stability: whether the same or a very similar 204 (or 20) proteins would be selected if Boruta were re-run on a bootstrap resample of the training data, or with a different random seed. I did not find a stability analysis of this kind reported. This matters because a single Boruta run, however well-conducted, does not by itself indicate how sensitive the selected protein set is to sampling variability — Extension 3 in Section 12 is built around this exact question.

Table 1. Evidence and interpretation.

Result	What it establishes	What it does not establish	Useful next test
High age-prediction correlation (UKB $r=0.94$; CKB $r=0.92$; FinnGen $r=0.94$)	Plasma proteins carry a strong, cross-population signal tracking chronological age	That the age-prediction residual (ProtAgeGap) reflects one coherent biological-aging process	Residual/calibration diagnostics across age strata
External age validation in CKB and FinnGen	The trained clock transports to genetically and geographically distinct samples for the age-prediction task	That disease and mortality associations transport equally well (most were tested only in UKB)	External replication of Cox associations in CKB/FinnGen or another cohort
ProtAgeGap-disease hazard ratios (e.g., AD HR 1.16, 95% CI 1.12–1.20)	A robust, adjusted, FDR-corrected association between ProtAgeGap and incident disease	That ProtAgeGap improves individual-level risk prediction beyond existing risk factors	Baseline vs. baseline+ProtAgeGap discrimination/calibration comparison

Result	What it establishes	What it does not establish	Useful next test
SHAP feature importance for 204 APs	Which proteins the fitted LightGBM model relies on most to predict age	That those proteins cause aging or the diseases ProtAgeGap predicts	Perturbation or Mendelian randomization evidence for top proteins
STRING high-confidence coexpression network (66-protein subnetwork)	That many of the 204 APs show high-confidence coexpression relationships in an external database	That this network reflects direct physical interaction, was experimentally validated within this study, or identifies mechanism	Targeted functional follow-up on hub proteins (e.g., EGFR, CXCL12)
Disease-free sensitivity analysis (n=20,315)	That most ProtAgeGap-phenotype associations are not purely driven by reverse causation from undetected disease	That all confounding (including residual, unmeasured confounding) is ruled out	Longitudinal repeat-measures analysis of ProtAgeGap trajectories before diagnosis

Each row separates what a specific reported result demonstrates from what it does not, and points to a test that would close the gap.

6. External Validation: Transportability of Age Prediction

This is, in my view, one of the paper's genuine strengths, and it deserves stating plainly before any caveats. The trained LightGBM/Boruta pipeline, fit entirely on UKB data, was applied without retraining to CKB and FinnGen and reproduced strong age-prediction accuracy in both: $r = 0.92$ ($R^2 = 0.82$, RMSE = 4.97, MAE = 3.90) in CKB, and $r = 0.94$ ($R^2 = 0.87$, RMSE = 5.33, MAE = 4.12) in FinnGen, compared with $r = 0.94$ ($R^2 = 0.88$, RMSE = 2.85, MAE = 2.24) in the UKB holdout test set. FinnGen extended the tested age range down to 19 years, roughly 20 years below the UKB's lower limit of 39, although mean ages were similar across the two cohorts (approximately 56 and 57 years, respectively). FinnGen participants were also largely free of major disease, so the model's performance there is a meaningfully different test than simply re-running it on more UKB-like data.

What this establishes is that the mapping from Olink-measured plasma proteins to chronological age, learned in one population, transports well to two other populations that differ in ancestry, geography, healthcare system, and (for FinnGen) age distribution and health status. That is a nontrivial result; many prediction models degrade substantially outside their training population, and this one largely did not, at least for the age-prediction task itself.

The distinction I want to keep sharp is between this specific claim — the clock's ability to predict chronological age was externally validated — and the much broader claim that the paper's disease and mortality findings were externally validated. They were not, for the most part. In CKB, the paper reports cumulative-incidence comparisons across ProtAgeGap deciles for diseases with more than ten incident cases across those deciles (including IHD, all-cause mortality, all stroke, ischemic stroke, type 2 diabetes, COPD, chronic liver disease, and CKD), with the authors themselves noting that confidence intervals were wide because of the smaller number of incident cases. This is a real, if more limited, form of external evidence — it is not the same as replicating the fully adjusted, FDR-corrected Cox proportional-hazards models run in the UKB. FinnGen was not used for outcome associations at all, given its near-absence of incident disease and only 22 recorded deaths. So while I would say the paper's central age-prediction claim is well externally validated, its disease-risk and mortality claims rest much more heavily on the single UKB sample, with CKB providing partial, descriptive support for a subset of outcomes.

	UKB	CKB	FinnGen
Sample size (n)	45,441	3,977	1,990
Study role	Discovery / development (training + holdout test)	External age-prediction validation	External age-prediction validation
Cohort design	Prospective general-population cohort	IHD case-cohort study	Mostly healthy blood donors
Age range (years)	39–71	30–78	19–78
% female	54.1%	53.7%	51.9%
Follow-up	11–16 years	11–14 years	Not applicable (age-prediction only)
Deaths / mortality	4,828 (10.6%)	1,426 (35.9%)	22 (1.1%)
What this cohort tests	Model development; nearly all outcome associations	Age-prediction transportability; descriptive disease-decile incidence only	Age-prediction transportability across a wider age range; largely disease-free blood donors

Figure 3. Cohort comparison across the three samples used in the study. Values verified against Table 1 and the Methods of Argenterio et al. (2024).

7. Disease Risk, Mortality, and Generalisability

Within the UKB, ProtAgeGap was associated with all 14 noncancer diseases studied except Parkinson's disease, and with all-cause mortality, across three progressively adjusted Cox models (age and sex; plus sociodemographic and lifestyle factors; plus BMI and prevalent hypertension). Effect sizes were largely stable across these adjustment levels, and the largest hazard ratios in the fully adjusted model were for Alzheimer's disease (HR 1.16), all-cause dementia (HR 1.12), and chronic kidney disease (HR 1.10), all per one-year increase in ProtAgeGap. ProtAgeGap20, built from just 20 proteins, was associated with every disease investigated, including Parkinson's disease. Of 12 cancers examined, cumulative-incidence differences by ProtAgeGap decile were observed for eight, and four remained significantly associated in adjusted Cox models: esophageal, lung, non-Hodgkin lymphoma, and prostate cancer.

Reverse causation — the concern that undiagnosed early-stage disease is itself changing protein levels, rather than proteomic aging predisposing to later disease — is a reasonable worry for any prospective biomarker study. The authors addressed this specifically for the aging-phenotype and biomarker associations via a disease-free sensitivity analysis (n = 20,315 UKB participants with no lifetime diagnosis of any of the 26 diseases studied at baseline); associations with nearly all markers persisted in this subsample, with a small number of specific exceptions (albumin, total bilirubin, self-rated facial aging, sleeping 10+ hours per day, and feeling tired every day dropped out of significance). Separately, for the Cox mortality and incident-disease models, the authors reported further sensitivity analyses restricted to never-smokers and to a normal BMI range; these showed broadly consistent hazard-ratio estimates, with Parkinson's disease being the outcome most often not reaching significance across these restricted samples. Reducing reverse-causation risk in the phenotype associations via a disease-free subsample, and checking the disease-incidence hazard ratios against two major confounders separately, is a more thorough workflow than a single unadjusted association would represent. It is still, importantly, association evidence — restricting to a disease-free-at-baseline sample reduces one specific threat to interpretation (reverse causation from an already-present disease) without eliminating residual confounding from factors not captured by the covariates included.

On generalisability, I think the most accurate summary is more differentiated than either ‘this clock works for everyone’ or ‘this clock only applies to one population.’ The UKB itself is a healthy-volunteer sample and predominantly of White ethnicity (93.1%), which is a real constraint on how far its disease-association findings generalize, independent of the external validation story. CKB's case-cohort design around IHD means its participants are not a simple population-representative sample of China, and its descriptive disease-decile findings (rather than fully adjusted Cox models) reflect that. FinnGen's proteomic subsample was deliberately drawn from largely healthy blood donors with a wider age range extending down to 19 years, which is precisely why it was suitable for testing whether the age-prediction model works across a broader span of adulthood but not suitable for testing mortality or disease associations. Comparable Olink Explore measurements were available across all three cohorts, which supports cross-cohort transportability of the age-prediction model, but sample processing, laboratory setting, and quality control were cohort-specific (CKB samples, for example, were split across two separate Olink laboratories), so this is not the same as independent cross-platform validation. I think ‘diverse populations’ in the title is a fair description of the age-prediction validation specifically; extending that same characterization to the disease-risk and clinical-utility claims would be over-reading what CKB and FinnGen individually contribute to those specific analyses.

8. Strengths and Limitations

8.1 Strengths

- Large, well-powered UKB development sample (n = 45,441) with a proper 70:30 train/holdout split.
- External validation of the age-prediction model in two cohorts genuinely distinct in ancestry, geography, and health status.
- Systematic benchmarking of six machine-learning approaches, with model selection based on external generalisability rather than in-sample accuracy.
- Broad, systematically tested outcome panel (27 aging phenotypes, 14 diseases, 12 cancers, mortality), with FDR correction across each family of tests.
- Disease-free, never-smoker, and normal-BMI sensitivity analyses addressing reverse causation and key confounders.
- A reduced, more translatable 20-protein model (ProtAge20) retaining most of the full model's age-prediction performance.
- Publicly available analysis code (R and Python) with a documented GitHub repository.
- Clear, if controlled-access, data pathways for each of the three cohorts used.

8.2 Limitations

Acknowledged by the authors: the model relies on the Olink Explore 3072 assay and therefore does not capture proteins outside that panel, including those covered by the larger Olink HT (~5,000 proteins) or SOMAscan (>10,000 proteins) platforms; and none of the three cohorts had DNA-methylation data available, so no direct within-study comparison between this proteomic clock and DNAm clocks was possible.

From my own reading: the training target is chronological age, which (as discussed in Section 4.1) does not by itself establish that ProtAgeGap reflects a single coherent biological-aging process; no head-to-head incremental-prediction comparison (discrimination, calibration, or reclassification) between a baseline clinical model and baseline-plus-ProtAgeGap is reported; the great majority of disease and mortality associations were tested only in the UKB rather than externally replicated; SHAP importance and the STRING coexpression

network establish model reliance and biological plausibility respectively, not causal mechanism; all three cohorts share the same underlying proteomic platform, so platform-independence has not been tested; I could not fully verify the exact nesting of Boruta feature selection within the cross-validation loop from the material reviewed, though this is a documentation gap rather than evidence of a problem; and no feature-selection stability analysis under resampling is reported for either the 204-protein or 20-protein models.

None of these amount to a fatal flaw in the study. They bound how far specific claims — particularly ‘biological aging’ and ‘diverse populations’ — can be pushed without overstating what the reported analyses directly demonstrate.

Table 2. Balanced evaluation.

Study feature	Strength	Limitation	Most useful follow-up
UKB sample size (n=45,441)	Large, well-powered for both model training and disease association testing	UKB is a healthy-volunteer cohort, predominantly White (93.1%), and not representative of the general population	Replicate key associations in a more representative or higher-risk sample
External cohorts (CKB, FinnGen)	Genuinely distinct ancestry, geography, and healthcare context for age-prediction validation	CKB is an IHD case-cohort sample, not population-representative; FinnGen had very few deaths (n=22) and was not used for most outcome associations	A prospective, population-based external cohort with adequate disease/death events for full Cox replication
Olink Explore 3072 platform	Comparable Olink Explore measurements available across all three cohorts, supporting cross-cohort transportability	Sample processing, lab site, and QC were cohort-specific (e.g., CKB run across two labs); covers 2,897 of many more circulating proteins; may not generalize to Olink HT or SOMAscan	Cross-platform comparison (e.g., against SOMAscan) of selected proteins and performance
LightGBM model choice	Selected for external generalisability rather than in-sample accuracy; outperformed linear and neural-network alternatives out-of-cohort	Tree-ensemble predictions are not directly interpretable without a post hoc attribution method	Formal comparison of SHAP-derived structure against an independent interpretable model class
Boruta / SHAP feature selection	Principled, permutation-based selection reduced 2,897 proteins to a selected and re-evaluated 204-protein set that retained comparable age-prediction performance	No reported stability analysis under resampling; exact nesting with cross-validation not fully verifiable from the material reviewed	Bootstrap-resampled Boruta runs to quantify selection-frequency stability
Outcome associations (27 phenotypes, 14 diseases, 12 cancers, mortality)	Broad, systematically tested, FDR-corrected, with disease-free and never-smoker sensitivity analyses	Nearly all tested only in the UKB; no head-to-head incremental-prediction comparison reported	Baseline-vs-baseline+ProtAgeGap discrimination/calibration analysis
Reduced 20-protein model (ProtAge20)	Retains ~95% of age-prediction performance with a far smaller, more translatable protein panel	Somewhat weaker for frailty/physiological associations than the full 204-protein model	Independent validation of ProtAge20 specifically (not just ProtAge) in a new cohort

Study feature	Strength	Limitation	Most useful follow-up
Code and data access	Public GitHub repository with R and Python analysis code	Cohort data require controlled access via each biobank's governance process, which limits exact independent reproduction	A fully containerized reproduction attempt using publicly documented environment details

Each study feature is paired with its principal strength, its principal limitation, and the follow-up analysis that would most directly address that limitation.

9. Reproducibility

R and Python code for data preparation and analysis is publicly available at the GitHub repository linked in the paper, and the Methods section documents software packages, key hyperparameter-search procedures (Optuna, with trial counts specified), and model configuration details (for example, LightGBM run with 5,000 estimators and 20 early-stopping rounds across each analysis step). This is a meaningfully more complete level of methodological documentation than a paper that simply reports final metrics.

It is worth distinguishing several things that 'reproducibility' can mean here. Code availability is satisfied: the pipeline logic is public. Data accessibility is a separate matter — UKB, CKB, and FinnGen data are all available only through each biobank's own controlled-access application procedure, not as an open download. I don't think this counts as a failure of open science; genomic and health-record data of this sensitivity generally cannot ethically be distributed without a governed access process, and the paper is transparent about how to apply for each. Computational reproducibility — whether an independent analyst with approved data access, using the public code, would recover the same 204-protein model and the same reported metrics — is harder for me to fully assess from the material reviewed; I did not find explicit documentation of random seeds for every stochastic step (for example, the Optuna hyperparameter searches and the fivefold cross-validation splits), so I can't be certain whether a rerun would reproduce the exact same 204 (or 20) selected proteins versus a broadly similar but not identical set. Exact independent reproduction and full external replication (an entirely separate research group repeating the analysis on new data and reporting whether the same associations hold) are different, stronger standards again, and this paper does not attempt the latter for itself.

The code is public, but exact reproduction still requires authorised access to the underlying cohort data and, from what I could verify, sufficiently complete documentation of the stochastic elements of the computational pipeline to guarantee an identical rerun rather than a closely similar one.

10. My Critical Perspective

What convinces me most about this paper is the combination of two things happening together: a large, well-powered development sample, and genuine external validation of the age-prediction task in two cohorts that are actually different from the UKB in ways that matter — different ancestry, different healthcare system, and in FinnGen's case a population with a wider age range, including younger adults, and consisting largely of blood donors. It would have been easy for a model tuned this closely to UKB protein distributions to fall apart outside the UKB, and it largely didn't for the age-prediction task. The sensitivity analyses made the epidemiological part of the study more careful than I initially expected.

What I remain uncertain about is narrower than a wholesale doubt about the clock's value. I don't think the paper establishes that ProtAgeGap measures one coherent, causally meaningful biological-aging process, as opposed to a useful but heterogeneous composite of age-associated protein signal. I also don't think it establishes, in the material I reviewed, that ProtAgeGap would materially improve an individual's risk

prediction beyond chronological age and existing risk factors, because the specific comparison that would show that — baseline model versus baseline-plus-ProtAgeGap, on a discrimination or calibration metric — is not reported. These two points matter because they sit right where the paper's own framing is most ambitious: the abstract's language about proteomic aging being 'used to predict' disease risk, and the Discussion's suggestion that the clock could identify 'protein targets for possible drug treatment,' both lean on exactly the kind of mechanistic and incremental-prediction evidence that I think is present here only in a partial, plausibility-level form.

This uncertainty is not a reason to dismiss the study. An adjusted, FDR-corrected, dose-graded, partly reverse-causation-controlled association between a biomarker and thirteen different disease outcomes plus mortality is a real and useful finding on its own, independent of whether it eventually supports a causal or clinically actionable claim. My concern is about which claims that finding directly supports, not about whether the finding itself is solid.

11. What Would Change My Mind

- A direct comparison of a baseline clinical risk model against the same model plus ProtAgeGap, reported on a discrimination metric (C-index or time-dependent AUC), a calibration assessment, and ideally a reclassification statistic, for at least one major outcome such as mortality or CKD.
- External replication of the fully adjusted Cox associations for mortality and the strongest disease signals (for example, CKD and dementia) in CKB, FinnGen, or another independent prospective cohort with adequate case numbers.
- A residual-versus-age or calibration-slope analysis showing ProtAgeGap behaves consistently across the full adult age range, sex, and ancestry groups, addressing the regression-to-the-mean question directly rather than through similar-distribution comparisons alone.
- Feature-selection stability evidence — for example, bootstrap-resampled Boruta runs showing the 204-protein (or 20-protein) set is not highly sensitive to sampling variability.
- Functional or causal evidence (Mendelian randomization, a perturbation experiment, or longitudinal within-person protein-change data) linking one or more of the top SHAP-ranked or STRING-hub proteins to a specific aging-related outcome, moving beyond association-based importance.

12. Potential Research Extensions

Extension 1 — Incremental clinical prediction

Limitation addressed: the paper does not report whether ProtAgeGap improves individual-level risk prediction beyond chronological age and established risk factors.

Research question: does adding ProtAgeGap to a baseline survival model containing chronological age, sex, and standard clinical risk factors improve discrimination or calibration for a major outcome such as all-cause mortality or CKD?

Proposed method: using UKB data (subject to the standard application process), fit a baseline Cox model with established covariates, then a second model adding ProtAgeGap, and compare C-index, time-dependent AUC, calibration slope/intercept, and, where appropriate, a decision-curve or net reclassification analysis between the two.

Evidence generated: a quantified estimate of incremental discrimination and calibration change attributable specifically to ProtAgeGap.

Interpretation of possible outcomes: a meaningful improvement in discrimination or calibration would directly support the clinical-utility framing in the paper's Discussion; a negligible improvement would indicate that ProtAgeGap's value is currently better understood as a research signal of biological aging than as a ready clinical risk-prediction tool, without implying the association itself was spurious.

Extension 2 — Age-gap calibration and residual diagnostics

Limitation addressed: no residual-versus-age plot or calibration-slope analysis was identified in the material reviewed, leaving the regression-to-the-mean question addressed only indirectly.

Research question: does ProtAgeGap behave consistently (mean near zero, similar spread) across chronological-age strata, sex, and cohort, or does it show systematic drift at the extremes of the age range?

Proposed method: plot ProtAgeGap (and ProtAge itself) against chronological age within each cohort; fit calibration intercept and slope; compare mean ProtAgeGap and its variance within five-year age bands; repeat separately by sex and cohort.

Evidence generated: a direct, visual and quantitative picture of whether the age-prediction residual is stable across the range it is meant to describe.

Interpretation of possible outcomes: a flat, centered residual across age strata would strengthen confidence that ProtAgeGap is not simply an artifact of regression to the mean; a systematic drift would identify a specific, correctable calibration issue rather than undermining the clock altogether.

Extension 3 — Protein-selection stability

Limitation addressed: no feature-selection stability analysis under resampling was identified for the 204-protein or 20-protein models.

Research question: how consistently are the same 204 (and 20) proteins selected when Boruta and the SHAP-based recursive elimination are re-run on bootstrap resamples or repeated cross-validation splits of the UKB training data?

Proposed method: repeat the Boruta selection procedure on, for example, 100 bootstrap resamples of the training set, recording each protein's selection frequency; similarly repeat the SHAP-based recursive elimination for the 20-protein reduction; compare selection-frequency distributions, SHAP-rank stability across runs, and resulting age-prediction performance.

Evidence generated: selection-frequency and rank-stability statistics for every candidate protein, plus a distribution of age-prediction performance across resampled models.

Interpretation of possible outcomes: high selection frequency for most of the reported 204 (or 20) proteins would support treating the published protein set as a stable signature; substantial instability would suggest the specific protein list is less important than the aggregate predictive signal, which would itself be a useful and publishable finding about how to interpret this and similar clocks.

13. Questions the Paper Leaves Open

- How much does ProtAgeGap improve individual-level risk prediction beyond chronological age and conventional clinical risk factors, measured on a discrimination or calibration metric rather than an adjusted hazard ratio?
- Are the mortality and disease associations reported in the UKB reproduced prospectively, using the same fully adjusted Cox approach, in CKB, FinnGen, or an independent cohort with adequate case numbers?
- How stable are the 204-protein and 20-protein selections under resampling, and would a different random seed or bootstrap sample select a meaningfully different protein set?
- Does ProtAgeGap remain well calibrated across the full adult lifespan, or does its behavior at the youngest and oldest ages depend on regression toward the mean?
- Would a proteomic clock built on a different platform (for example, SOMAscan or the larger Olink HT panel) select a similar set of age-predictive proteins and achieve comparable performance, or does platform choice materially change which proteins appear important?

14. Lessons Learned

First, a model that predicts chronological age very accurately has not, by that fact alone, validated a measure of biological age. I came into this paper conflating the two more than I realized, and separating ‘the model recovers age well’ from ‘the residual means something biologically coherent’ is now something I check for explicitly, especially because the second claim is the one that gets used in the framing and Discussion even when the direct evidence supports something closer to the first.

Second, an adjusted, statistically significant hazard ratio answers a different question from an incremental-prediction comparison, and a paper can report a great deal of the former without directly reporting the latter. Before working through this paper's Cox models carefully, I think I would have read ‘ProtAgeGap was significantly associated with 13 of 14 diseases’ as close to ‘ProtAgeGap improves disease prediction,’ which is not the same claim, and I now want to look specifically for a baseline-versus-baseline-plus-marker comparison before accepting the stronger version.

Third, feature-importance tools like SHAP, and even external database checks like a STRING coexpression network, describe a model or a coexpression pattern — they are not a substitute for perturbation or causal evidence, however well they are presented. I found it easy to read a well-constructed protein-interaction figure as more mechanistically conclusive than the underlying analysis actually supports, and I think that is a general risk with model-interpretation tools that produce visually compelling network diagrams.

15. Connection to My Development as a Bioinformatician

This paper sharpened a specific list of technical skills I want to keep building rather than a general sense that proteomics is interesting. Working through the Methods gave me a much more concrete picture of what a full machine-learning pipeline for a biomarker paper actually involves: benchmarking multiple model classes rather than assuming the most flexible one wins, choosing a final model on external generalisability rather than in-sample accuracy, running hyperparameter tuning nested inside cross-validation, and layering a principled feature-selection method (Boruta) with a post hoc interpretation method (SHAP) on top of the prediction task rather than treating either as a replacement for the other.

Specific skills I want to develop further, prompted directly by this paper, include: designing train/holdout/external-validation splits that actually separate model development from generalisability testing; running and interpreting SHAP values correctly, including the distinction between feature importance and feature interaction; building and critically reading Cox proportional-hazards models, including the difference between adjusted association and incremental discrimination; assessing model calibration rather than only discrimination; and writing reproducible Python and R workflows with the kind of documentation (software versions, tuning procedures, hyperparameter search budgets) that this paper's Methods section models reasonably well.

The realistic project I would like to attempt, building on Extension 1 above, is to reconstruct a small, openly reproducible version of a biomarker-based aging or risk model using an accessible public dataset, train it to predict a simple, well-defined target the way ProtAge was trained on chronological age, and then directly compare that baseline target-prediction accuracy against the model's incremental value for predicting one downstream health outcome, using a discrimination or calibration metric rather than an association statistic alone. I have not implemented LightGBM, SHAP, Boruta, or Cox modeling in a research project of my own yet; this paper gave me a clearer sense of what the full pipeline should look like and which comparisons actually matter, which feels like the right place to be starting from before attempting the project itself.

16. Conclusion

This study convincingly establishes that plasma protein levels, measured on a single consistent platform, can predict chronological age with high accuracy in the UK Biobank and that this accuracy transports well to two independent cohorts distinct in ancestry, geography, and health status. It further establishes, through a large, FDR-corrected, sensitivity-tested set of Cox and regression models within the UKB, that the resulting age-prediction residual (ProtAgeGap) is robustly associated with a wide range of aging-related phenotypes, incident diseases, cancers, multimorbidity, and all-cause mortality, independent of chronological age and standard covariates.

It suggests, but does not directly establish from the material reviewed, that ProtAgeGap would materially improve individual-level clinical risk prediction beyond existing risk factors, since no baseline-versus-baseline-plus-ProtAgeGap discrimination or calibration comparison is reported; that ProtAgeGap represents one specific, causally coherent biological-aging mechanism, since the supporting evidence (SHAP importance, the STRING coexpression network) speaks to model reliance and biological plausibility rather than demonstrated mechanism; and that the paper's disease and mortality findings, as opposed to its age-prediction findings specifically, have been validated across genuinely diverse populations, since most outcome analyses were conducted in the UKB alone.

The strongest part of the design is the combination of a large, properly held-out UKB development sample with real external validation of the age-prediction task in two cohorts that differ from the UKB in ways that plausibly could have broken the model but did not. The most important interpretive limitation is the gap between the extensive adjusted-association evidence the paper reports and the incremental-prediction evidence that a stronger clinical-utility claim would require. The most useful next analysis, in my view, is a direct comparison of a standard clinical risk model with and without ProtAgeGap on a discrimination and calibration metric, since this single comparison would do more than any other single addition to clarify how far the clock's genuinely strong association evidence extends toward the clinical-prediction and mechanistic claims the paper's framing sometimes gestures toward.

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